



UNIVERSITÄT
LEIPZIG

Asymmetric Numeral System

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Introduction

Introduction

- Hi there, this is a self inspired latex beamer template for Leipzig university.

Entropy Coding

1st frame

- This is one of the section¹.
- Let's check for 2nd footnote ².
- The footnotes will be added from the bottom. If you want to pull the footnotes towards red-black margin line then use `\addfootspace{}` at the end of the `\frame{}`.
- `\addfootspace{}` actually adds an additional vacant footnote (without counting that vacant footnote).
- To add more foot space add more `\addfootspace{}`.

¹This is a long footnote. Here I am. Who am I?

²This is a 2nd long footnote. Here I am. Who am I?

2nd frame

- Here is the next section³
- Here I used two `\addfootspace{}`. Don't do `\addfootspace{\vspace{1cm}}`.

³This is a 2nd long footnote. Here I am. Who am I?

Asymmetric Numeral System

Motivation

ANS combines the compression ratio of arithmetic coding (which uses a nearly accurate probability distribution), with a processing cost similar to that of Huffman coding [3]

Basic Idea

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- rANS and tANS

Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]

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Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]
- $\bar{s} : \mathbb{N} \rightarrow \mathcal{A}, \bar{s}(x) = \text{mod}(x, b) = s$
- In the standard binary system example: $\bar{s}(x) = \text{mod}(x, 2)$
e.g. splits natural numbers into subsets of even/odd numbers [1]

Symbol spread function (Exa. 1)

$$S = \{0, 1\}$$

Exa. 1: Standard Binary System

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
s=0	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow
s=1	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White	Yellow	White

Fig.1

Symbol spread function (Exa. 2)

$$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

Exa. 2: Standard Decimal System

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
s=0																					
s=1																					
s=2																					
s=3																					
s=4																					
s=5																					
s=6																					
s=7																					
s=8																					
s=9																					

Fig.2

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

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- How can we represent this sequence as a single number?
- The easiest way is to concatenate the symbols/digits:

$$X_{out} = 2025$$

Representation of Information (Exa. 2)

- encoding the sequence of digits (2,0,2,5,1,8)

$$X_{\text{init}} = 0$$

$$X_1 = X_{\text{init}} \times 10 + 2 = 2$$

$$X_2 = X_1 \times 10 + 0 = 20$$

$$X_3 = X_2 \times 10 + 2 = 202$$

$$X_{\text{out}} = X_3 \times 10 + 5 = 2025$$

Representation of Information (Exa. 2)

- What is the encoding function?

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- x-th appearance of symbol s

$$X' = C(s, X) = X \cdot 10 + s$$

Representation of Information (Exa. 2)

Exa. 2: Standard Decimal System

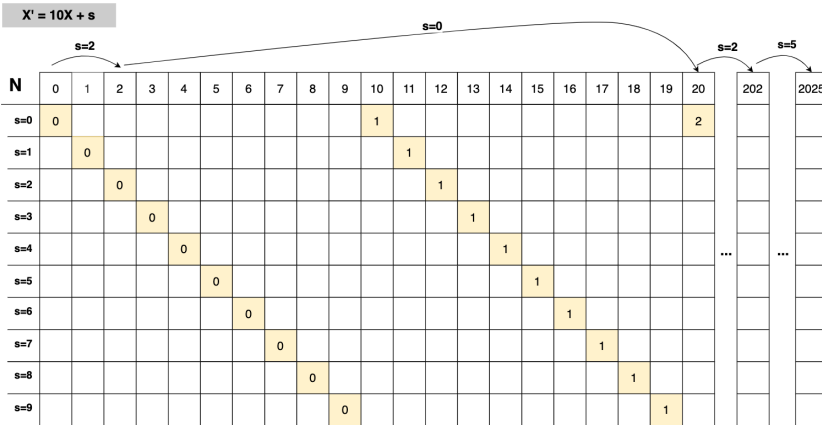


Fig.3

Representation of Information (Exa. 1)

Exa. 1: Standard Binary System

$X' = 2X + s$

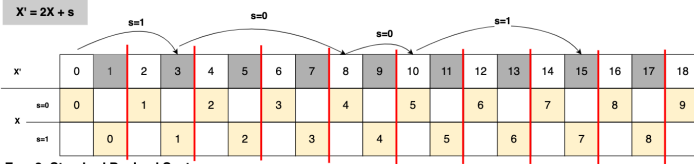
$s=1$ $s=0$ $s=0$ $s=1$

x'	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
x $s=0$	0		1		2		3		4		5		6		7		8		9
$s=1$		0		1		2		3		4		5		6		7		8	

Fig.4

Asymmetric Numeral System | Basic Idea | Symmetric or Asymmetric

Exa. 1: Standard Binary System



Exa. 2: Standard Decimal System

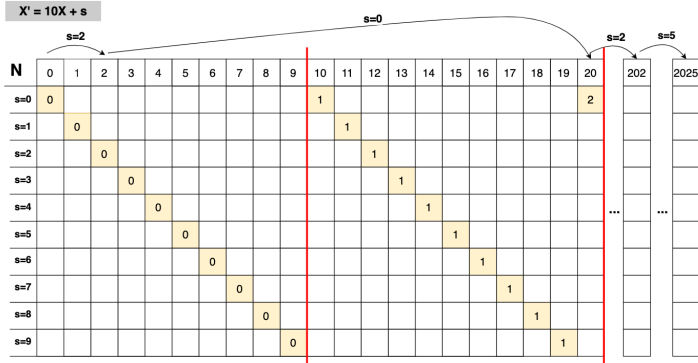


Fig.6

Symmetric or Asymmetric

- Symmetric: uniform distribution
- Asymmetric: non-uniform distribution

Exa. 3

- Let $\mathcal{A} = \{0, 1\}$
- $Pr(0) = 1/4, Pr(1) = 3/4$
- a cycle of length 4, which contains 1 zero and 3 ones
- $X' = C(s, X) \approx X/p_s$
- $C(0, X) = 4X, C(1, X) = 4\lfloor X/3 \rfloor + \text{mod}(X, 3) + 1$

rANS

- range

-

Encoding Formula:

$$x' = \left\lfloor \frac{x}{F_s} \right\rfloor \cdot m + \text{mod}(x, F_s) + c_s$$

- **Alphabet set:** $\mathcal{A} = \{a_1, a_2, \dots, a_k\}$
- **Absolute frequency of symbol s :** F_s
- **Total frequency:**

$$m = \sum_{i=1}^k F_i$$

- **Probability of symbol s :**

$$p_s = \frac{F_s}{m}$$

- **Cumulative frequency before s :**

tANS

—

Performance

—

Summary

Summary

- To summarize, this may help you to make a Leipzig university beamer presentation.

Reference

Reference

1. J. Duda, K. Tahboub, N. J. Gadgil and E. J. Delp, "The use of asymmetric numeral systems as an accurate replacement for Huffman coding," 2015 Picture Coding Symposium (PCS), Cairns, QLD, Australia, 2015, pp. 65-69, doi: 10.1109/PCS.2015.7170048.
2. 2
3. https://en.wikipedia.org/wiki/Asymmetric_numeral_systems